SE LAB 1

Micro-Lending & Peer Credit Risk Assessor

PROBLEM STATEMENT #33

Name: Kavan S
SRN: PES2UG24CS223
Domain: Retail, E-Commerce & Finance

Lab 1:

1. **Requirements Table**

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	**Functional**	**The system shall calculate the borrower's credit-risk rating based on debt-to-income ratio, repayment history, and employment stability.**	**High**	**Pass: Appropriate Low, Medium, or High risk tier is assigned with complete information. Fail: Risk rating is assigned with incomplete required information.**	**Credit-risk assessment helps peer lenders evaluate borrowers.**
FR-002	**Functional**	**The system shall allow a Borrower to submit a micro-loan request by providing the required loan amount and repayment period**	**High**	**Pass: Valid loan amount and repayment period create a unique loan request ID. Fail: Request is created with missing or invalid mandatory information.**	**Allows borrowers to initiate the peer-to-peer lending process.**
FR-003	**Functional**	**The system shall generate a structured repayment schedule for an approved**	**High**	**Pass: Approved loan produces repayment dates and amounts.**	**Provides borrowers and lenders with a clear repayment plan.**

		micro-loan based on the loan amount, repayment period, and applicable repayment terms.		**Fail: Approved loan does not have a repayment schedule.**	
FR-004	**Functional**	**The system shall allow a Peer Lender to review eligible loan requests and provide funding for a selected loan request.**	**High**	**Pass: Confirmed funding is recorded and loan status is updated. Fail: Funding is recorded without valid lender confirmation.**	**Enables peer lenders to make lending decisions and fund suitable borrowers.**
FR-005	**Functional**	**The system shall trigger an automated warning when a borrower fails to make a scheduled repayment by the required due time.**	**High**	**Pass: Warning is generated when repayment becomes overdue. Fail: No warning is generated for an overdue repayment.**	**Helps identify potential defaults and allows timely action.**
NFR-001	**Nonfunctional**	**The system shall execute loan repayment and EMI distribution operations within ACID-compliant database transactions to prevent double-spending and inconsistent financial records.**	**High**	**Pass: Transactions are atomic and do not create duplicate or inconsistent records. Fail: A failed transaction partially updates financial records.**	**Protects the integrity and consistency of financial transactions.**
NFR-002	**Nonfunctional**	**The system shall complete a**	**High**	**Pass: At least 95% of assessments**	**Ensures borrowers and peer**

		credit-risk assessment within 5 seconds under the specified simulated peak load.		**complete within 5 seconds. Fail: More than 5% exceed 5 seconds.**	**lenders receive risk information quickly.**

Summary

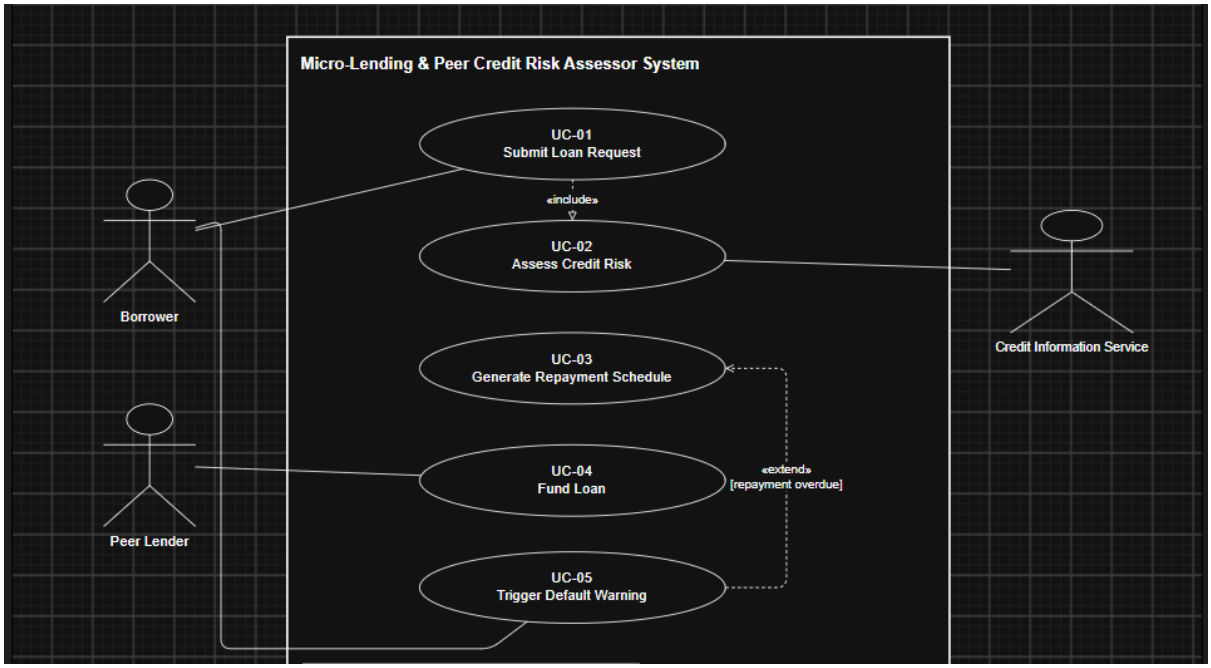
Functional Requirements: 5

- FR-001 — Calculate Credit Risk
- FR-002 — Submit Loan Request
- FR-003 — Generate Repayment Schedule
- FR-004 — Fund Loan
- FR-005 — Trigger Default Warning

Non-Functional Requirements: 2

- NFR-001 — ACID Transaction Integrity
- NFR-002 — Credit-Risk Assessment Performance

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-01 — Submit Loan Request

1. Preconditions

1. The Borrower is registered in the system.
2. The Borrower is logged into the system.
3. The Borrower's required financial information is available in the system.

2. Main Success Scenario

1. The Borrower selects **"Submit Loan Request"**.
2. The system displays the loan request form.
3. The Borrower enters the required loan amount and repayment period.
4. The system validates the entered loan information.
5. The system retrieves the Borrower's required financial information.
6. The system calculates the Borrower's credit-risk rating using debt-to-income ratio, repayment history, and employment stability.
7. The system assigns an appropriate **Low, Medium, or High** risk tier.
8. The system creates the loan request and generates a unique request ID.
9. The system displays the submitted loan request and credit-risk rating to the Borrower.
10. The use case ends successfully.

3. Alternate Flow — Invalid Loan Information

4a. Invalid loan information

1. The system detects that the loan amount or repayment period is missing or invalid.
2. The system displays an appropriate error message.
3. The Borrower corrects the invalid information.
4. The system validates the information again.
5. If the information is valid, the flow continues from Step 5 of the Main Success Scenario.

4. Postconditions

1. A valid loan request is stored in the system.
2. A credit-risk rating is associated with the loan request.
3. A unique loan request ID is generated.
4. The loan request is available for further processing by the Peer Lender.